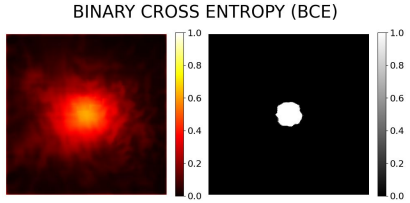
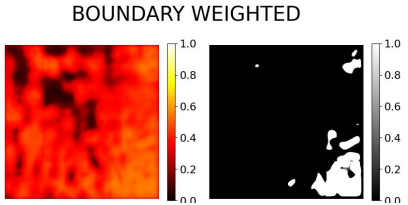
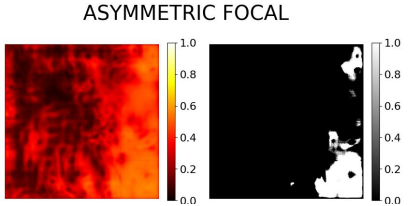
(A)



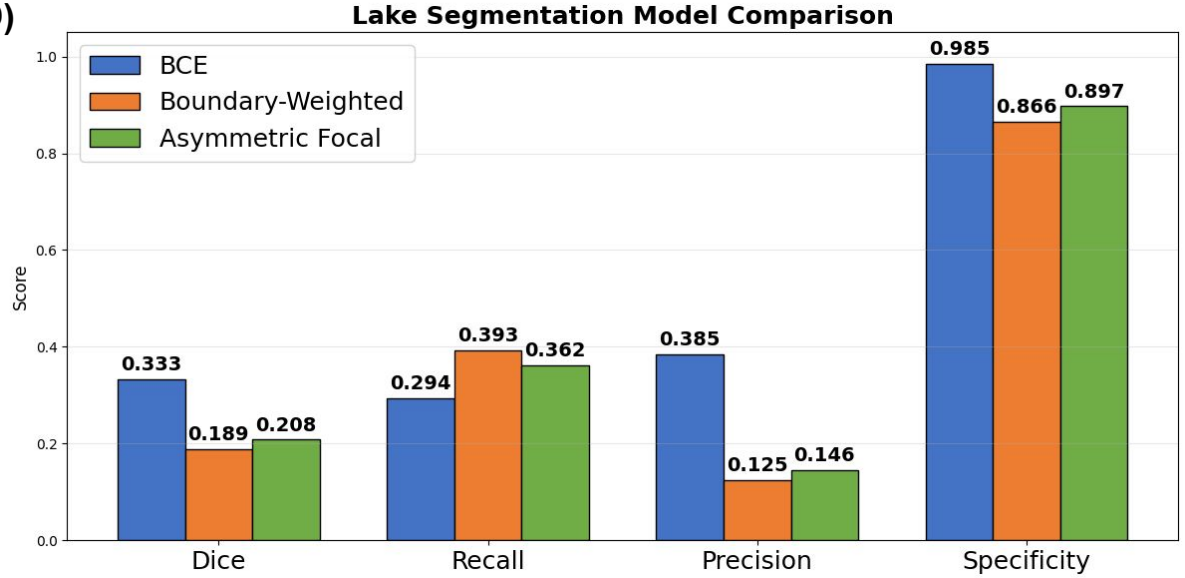

(B)




(C)




(D)




(E)

| Model | Dice | IoU | Precision | Recall | Specificity | Accuracy | FPR |
|---|---|---|---|---|---|---|---|
| BCE | 0.333 | 0.2 | 0.385 | 0.294 | 0.985 | 0.965 | 0.015 |
| Boundary | 0.189 | 0.104 | 0.125 | 0.393 | 0.866 | 0.844 | 0.134 |
| Asymmetric Focal | 0.208 | 0.116 | 0.146 | 0.362 | 0.897 | 0.872 | 0.103 |

**Figure 2.** Performance comparison of the three models we used on our problem. The first architecture had a loss function based solely on binary cross entropy, which equally weighted the accuracy of each pixel's prediction. The outcome of this model when applied to a null, lake-less image is shown in Panel (A). The left side of the figure is a probability map, where brighter pixels were predicted to have a higher probability of representing lakes. The right side of the figure is the final mask predicted by the model, where all pixels with a probability greater than 0.5 are marked as containing a lake and represented in white. This early draft of the model did not augment images, so the model overfit the pattern of lakes being centered in the input images. The second version of the model de-emphasized pixel-by-pixel accuracy, instead focusing on raising the dice score, which is a measure of how well the predicted mask overlaps with the true mask. The loss function for this model also dynamically adjusted weights to place less emphasis on boundary pixels, since we were more concerned with finding the bulk of the lake. We used image augmentation including random flips, rotations, and cropping to prevent overfitting. We also used image patching, by breaking up the images into subsections, passing them through the model, and then reconstructing the output into a full image. This resulted in some strange edge effects, as seen in the model output in Panel (B). The boundary-weighted model was applied to the same null image as in Panel (A), and there is a visible uptick in inaccuracies along the bottom and right edges. The third model used asymmetric focal loss, a method that places more emphasis on correctly identifying lake pixels, because they are a rare class that make up less than 5% of our data. This loss function placed less emphasis on the easy to classify background pixels, making the training process focus more on correctly identifying lake pixels. Combined with image augmentation, this loss function was effective. Panel (C) shows this asymmetric focal model applied to the same null mask, which again shows weird edge effects in the bottom right that are connected to our image patching method. Panel (D) explores metrics used to asses the models' performances. Although the binary cross entropy model (shown in Panel (A)) seems to outperform the others, it is overfitting our data. It would not be generalizable to other data sets. Panel (E) further explores performance metrics. For each column, the green represents the best performance, red represents the worst performance, and yellow is the performance in the middle. If we throw out the binary cross entropy model due to overfitting, we can compare the remaining two. The asymmetric focal model (shown in Panel (C)) outperforms the boundary-weighted model (shown in Panel (B)) in every aspect except recall, a ratio of the predicted positives out of the total number of actual positives. However, since the boundary-weighted model still has a worse false positive rate (FPR), the asymmetric focal model seems preferable for finding the most lakes with the highest accuracy.